

RESEARCH ARTICLE | AUGUST 19 2026

Neural wavefunction calculations of hyperfine coupling constants relevant to μ SR spectroscopy with quantum muons and protons

Special Collection: [Machine Learning for Electronic Structure](#)

Jamie Carr  ; Mathias Volkai  ; W. M. C. Foulkes  ; Andres Perez Fadon  



J. Chem. Phys. 165, 074114 (2026)

<https://doi.org/10.1063/5.0334570>



Articles You May Be Interested In

Highly accurate real-space electron densities with neural networks

J. Chem. Phys. (January 2025)

Partitioning the electronic wave function using deep variational Monte Carlo

J. Chem. Phys. (August 2025)

Two-component density functional theory for muonic molecules: Inclusion of the electron–positive muon correlation functional

J. Chem. Phys. (January 2022)

23 August 2026 20:47:38



The Journal of Chemical Physics

Special Topics Open for Submissions

[Learn More](#)

Neural wavefunction calculations of hyperfine coupling constants relevant to μ SR spectroscopy with quantum muons and protons

Cite as: J. Chem. Phys. 165, 074114 (2026); doi: 10.1063/5.0334570

Submitted: 18 March 2026 • Accepted: 4 August 2026 •

Published Online: 19 August 2026



Jamie Carr, , Mathias Volkai, , W. M. C. Foulkes, , and Andres Perez Fadon^{a)}

AFFILIATIONS

Department of Physics, Imperial College London, London SW7 2AZ, United Kingdom

Note: This paper is part of the Special Topic Machine Learning for Electronic Structure.

^{a)} Author to whom correspondence should be addressed: andres.perez-fadon19@imperial.ac.uk

ABSTRACT

Accurate prediction of muon hyperfine constants at zero temperature is useful for interpreting muon spin spectroscopy data, yet standard methods, such as density functional theory (DFT), evaluate these constants within the Born–Oppenheimer approximation, in which the electronic wavefunction is calculated with the muon and nuclei held as clamped point charges. This treatment may be inadequate for the muon because of its small mass. To overcome this limitation, we use the variational quantum Monte Carlo method with neural-network trial wavefunctions, a highly accurate and flexible approach recently applied to other quantum chemical problems. Within this framework, the muon can either be clamped or included explicitly in the many-particle electron-muon wavefunction, in which case the fully quantum-mechanical pair density is obtained directly. We calculate zero-temperature vacuum muon hyperfine constants in muoniated methyl and ethyl radicals for both clamped and quantum-mechanical muons. The hyperfine constants obtained from our clamped-muon calculations in the methyl and ethyl radicals differ significantly from the corresponding DFT results, highlighting the limitations of DFT even when the muon is clamped. The results with quantum muons are closer to experiment after accounting for environmental and temperature effects. These findings suggest that explicitly calculating the quantum-mechanical muon-electron pair density improves the accuracy of muon hyperfine constant predictions.

© 2026 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.1063/5.0334570>

I. INTRODUCTION

In a typical μ SR experiment, low-energy spin-polarized anti-muons (referred to as “muons” in most of the literature and throughout the rest of this paper) of charge $+e$ are implanted into solids and molecules, where they act as sensitive probes of the local magnetic environment.^{1–3} The angular distribution of the positrons produced by the resulting muon decay reveals details about the local spin density of materials.^{1,3} Frequently, the implanted muon interacts with an electron, causing the many-particle wavefunction to develop a muonium-like component: the muon binds to an electron to produce a muonium (Mu) atom, chemically analogous to a light hydrogen atom.^{1,4} When a muonium atom binds to an unsaturated closed-shell molecule, the result is a muoniated radical with an unpaired electron and a non-zero electron as well as

muon spin densities.^{2,4,5} For such radicals, key measurable quantities are the muon hyperfine constants, which represent the strength of the hyperfine interaction between the muon and electron spins.^{1–4} Computational predictions of these muon hyperfine constants can then be compared with experimental μ SR spectra to identify which chemical species are present in the sample.⁶

Hyperfine constants can be measured using transverse field μ SR (TF- μ SR).^{2,4,7,8} In an applied transverse magnetic field, the muon and unpaired electron in a muoniated radical form a superposition of four spin eigenstates: $|\uparrow_e\uparrow_\mu\rangle$, $|\uparrow_e\downarrow_\mu\rangle$, $|\downarrow_e\uparrow_\mu\rangle$, and $|\downarrow_e\downarrow_\mu\rangle$.^{1,4} As the magnetic moment of the electron is much larger than that of the muon, the Zeeman interaction separates these into electron-up and electron-down pairs, while the muon Zeeman and hyperfine interactions determine the energy splitting between the muon spin states within each pair.^{1,4} These splittings manifest themselves in

the Larmor precession of the muon spin as two distinct frequencies, which can be measured from the angular distribution of the decay positrons, emitted preferentially along the muon spin direction.^{1,4} As the hyperfine energy shift arises from the muon and the electron spins from being aligned or anti-aligned with the external magnetic field, the difference between the two precession frequencies cancels the muon Zeeman term and yields the isotropic muon hyperfine constant, A_μ .^{1,4}

Computational quantum chemistry methods are often used to calculate muon hyperfine constants, with the objective of replicating experimental results by including thermal and environmental effects.^{1,6,9–11} While high-accuracy methods, such as coupled cluster, can describe the ground state reliably, their computational cost is often prohibitive for large systems.¹² Density functional theory (DFT) provides a much faster alternative and, although generally less accurate, yields surprisingly good results.^{10,13}

The isotropic muon hyperfine constant depends on the electron spin density at the location of the muon,¹ obtained by taking the zero-separation limit of the muon–electron pair density function (PDF). This dependence exposes a key limitation of DFT methods. Since DFT is a mean-field-like theory that produces one-electron wavefunctions, two-particle correlation functions such as the pair density are not directly available.¹⁴

One solution to this problem is to adopt the simplest Born–Oppenheimer treatment, clamping the muon at a fixed position and replacing the muon–electron pair density by the electron density at the site of the clamped muon.^{9,10} The clamped-muon approach is also used in most beyond-DFT quantum chemical calculations and the results are often good. However, the mass of the muon is only 207 times that of an electron (about one ninth that of a proton), and the accuracy of this clamped-muon treatment is far from guaranteed. A fuller Born–Oppenheimer treatment retains the electronic energy as a function of the muon position, defining a potential energy surface on which the muon moves and acquires a wavefunction of its own. Earlier computational work in μ SR-related chemistry followed this route through vibrational averaging, zero-point energy corrections, and related effective-geometry models.^{15–19} For example, Böhm, Ramírez, and Schulte¹⁷ used Feynman path-integral quantum Monte Carlo to study zero-point and thermal averaging in the muoniated ethyl radical, including low-temperature estimates of the muon hyperfine constant. Approaches such as path-integral molecular dynamics have also been used successfully within a DFT framework to incorporate nuclear quantum and thermal effects in muoniated radicals,¹⁰ but in any such Born–Oppenheimer scheme, the muon–electron pair density is only ever evaluated with the muon held fixed. Averaging these clamped pair densities over the positional distribution of the muon is not equivalent to the pair density of a wavefunction in which the muon and electrons are explicitly correlated. Similarly, two-component DFT, often used to study electron–positron systems, is able to treat both electrons and muons quantum mechanically and can compute the electron and muon densities, but remains unable to describe electron–muon correlations.²⁰ The zero-separation limit of the electron–muon pair density is written as the product of the electron and muon densities multiplied by a correlation factor that must be approximated.²¹ Local-density-like approximations, in which the correlation factor is assumed to depend on the electron

and muon densities (and perhaps their gradients) at the coincidence point, have been used to compute μ SR frequencies in some cases.²²

Over the past few years, the neural wavefunction approach has shown itself able to describe molecules without muons at a level of accuracy comparable with high-level quantum chemistry methods such as coupled cluster with singles, doubles, and perturbative triples [CCSD(T)], while scaling more favorably to large system sizes.^{12,23–26} Furthermore, neural-network-based wavefunctions have been used to study quantum systems with multiple species of particles, such as positrons in molecules,²⁷ and dense hydrogen with quantum protons.²⁸ In this paper, we treat electron–muon correlations quantum mechanically by constructing a neural-network-based many-particle wavefunction ansatz that depends on the positions of electrons and muons, treating both on exactly the same footing. Through this formulation, the zero-point contribution of the quantum particles that would conventionally be treated through separate vibrational corrections is instead incorporated directly into the Coulombic many-body wavefunction. Our objective is to predict gas-phase muon hyperfine constants at zero temperature for muoniated methyl and ethyl radicals under both clamped and quantum treatments. By comparing the predicted muon hyperfine constants with the lowest-temperature available experimental data, for which thermal effects should be smallest, we show that a fully quantum treatment of the muon–electron system yields values closer to experiment, although residual condensed-phase anisotropic effects present in the experimental data are not included in the present model. Furthermore, when accounting for systematic environmental effects, our approach produces results that are more accurate than standard DFT calculations. We hope that in the future neural wavefunction methods will become an essential part of the computational toolkit used to interpret μ SR experiments.

A. Theory

We consider a chemical system described to a first approximation by the many-body Coulomb Hamiltonian in the Born–Oppenheimer clamped-nuclei approximation,

$$\hat{\mathcal{H}}_{\text{Coulomb}} = -\sum_i \frac{\hbar^2}{2m_i} \nabla_i^2 + \sum_{ij} \frac{q_i Q_j}{4\pi\epsilon_0 |\hat{\mathbf{r}}_i - \mathbf{R}_j|} + \sum_{i>j} \frac{q_i q_j}{4\pi\epsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|} + \sum_{i>j} \frac{Q_i Q_j}{4\pi\epsilon_0 |\mathbf{R}_i - \mathbf{R}_j|}, \quad (1)$$

where q_i , m_i , and $\hat{\mathbf{r}}_i$ denote the charges, masses, and position operators of the muon and electrons, respectively, while Q_i and $\mathbf{R}_i$ are the charges and positions of the clamped nuclei, respectively.

In addition to the Coulomb interaction, the full Hamiltonian includes hyperfine interactions between the spins of electrons and muons.¹ For rapidly reorienting molecular systems where thermal effects come into play, such as ones in the gas-phase, the anisotropic hyperfine interaction is motionally averaged, so here, we only consider the isotropic component, also known as the Fermi contact interaction.²⁹ The isotropic hyperfine interaction for many electrons and a single muon can be written as

$$\hat{\mathcal{H}}_{\text{hyperfine}} = \frac{2\mu_0}{3} \gamma_e \gamma_\mu \hbar^2 \hat{\mathbf{I}}_\mu \cdot \sum_i^N \hat{\mathbf{S}}_i \delta(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_\mu), \quad (2)$$

where $\hat{\mathbf{I}}_\mu$ is the muon spin operator, $\hat{\mathbf{S}}_i$ is the spin operator for the i th electron, and $\hat{\mathbf{r}}_\mu$ and $\hat{\mathbf{r}}_i$ are the corresponding position operators.³⁰ The constant μ_0 is the vacuum permeability, and γ_e and γ_μ are the electron and muon gyromagnetic ratios, respectively.

The ground state of the system with a muon and N electrons is described by the many-body wavefunction,

$$\Psi = \Psi(\mathbf{r}_\mu, \sigma_\mu; \mathbf{r}_1, \sigma_1; \dots; \mathbf{r}_N, \sigma_N), \quad (3)$$

which depends on the spatial coordinates $\mathbf{r}$ and spin coordinates σ of the muon and all of the electrons. It is useful to define the spin-resolved pair density function,

$$\rho_{\sigma_\mu \sigma_e}(\mathbf{r}_\mu, \mathbf{r}_e) = \sum_{\{\sigma\}} \int d\mathbf{R}_e \sum_{i=1}^N \delta(\mathbf{r}_e - \mathbf{r}_i) \delta_{\sigma_e \sigma_i} |\Psi|^2, \quad (4)$$

where σ_μ and σ_e are the muon and electron spin indices, respectively, $d\mathbf{R}_e \equiv d^3r_1, \dots, d^3r_N$, and $\{\sigma\} \equiv \{\sigma_1, \dots, \sigma_N\}$. The spin-resolved system-averaged pair density function is then defined as

$$\bar{\rho}_{\sigma_\mu \sigma_e}(\mathbf{u}) = \int d^3r_\mu \rho_{\sigma_\mu \sigma_e}(\mathbf{r}_\mu, \mathbf{r}_\mu + \mathbf{u}), \quad (5)$$

where $\mathbf{u} = \mathbf{r}_e - \mathbf{r}_\mu$ is the separation vector between the muon and the electron.

Recall the definition of the isotropic muon hyperfine constant A_μ (henceforth, just “hyperfine constant”),

$$\langle \hat{\mathcal{H}}_{\text{hyperfine}} \rangle = A_\mu \langle \hat{\mathbf{I}}_\mu \cdot \hat{\mathbf{S}} \rangle, \quad (6)$$

where $\hat{\mathbf{S}} = \sum_i \hat{\mathbf{S}}_i$ is the total electron spin operator.^{1,31} Placing all particles in σ_z eigenstates, parallel or anti-parallel to the applied magnetic field, and comparing the definitions of the Fermi contact Hamiltonian and the spin-resolved system-averaged pair density, we find that

$$A_\mu = \frac{2\mu_0}{3} \gamma_e \gamma_\mu \hbar^2 \Delta\rho(0), \quad (7)$$

where we have defined

$$\Delta\rho(\mathbf{r}) = \bar{\rho}_{\uparrow\uparrow}(\mathbf{r}) - \bar{\rho}_{\uparrow\downarrow}(\mathbf{r}) \quad (8)$$

and assumed, without loss of generality, that the muon is spin up.

II. METHOD

Evaluating the hyperfine constant requires finding the ground state of the full Hamiltonian, $\hat{\mathcal{H}} = \hat{\mathcal{H}}_{\text{Coulomb}} + \hat{\mathcal{H}}_{\text{hyperfine}}$, a task simplified by noting that the Coulomb interaction is several orders of magnitude stronger than the hyperfine interaction. For example, in muonium, the Coulomb ground state energy is 13.54 eV and the hyperfine constant is 18.5×10^{-6} eV—the latter is $\sim 700\,000$ times smaller than the former.¹ As such, first-order perturbation theory can be employed and we need only find the ground state of the Coulombic system, $\hat{\mathcal{H}} = \hat{\mathcal{H}}_{\text{Coulomb}}$. We use Hartree atomic units, with energies measured in Hartrees (E_h) and lengths in Bohr radii (a_0).

The variational Monte Carlo (VMC) method uses a parameterized trial ground-state wavefunction, $\Psi(\mathbf{R}; \theta)$, where $\mathbf{R}$ is shorthand for the list of all particle position coordinates, $\mathbf{R} := (\mathbf{r}_\mu, \mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$, and θ represents the adjustable parameters. As is always possible when the Hamiltonian is independent of spin, we have chosen to assign the spins,³² setting the muon spin up, the first $N^\uparrow$ electrons spin up, and the remaining $N^\downarrow = N - N^\uparrow$ electrons spin down. The wavefunction may then be viewed as a function of the spatial coordinates only. The parameters θ are optimized using a gradient descent procedure employing the Kronecker factored approximate curvature (KFAC) algorithm³³ to minimize the energy expectation value,

$$\langle \hat{\mathcal{H}} \rangle_\theta = \frac{\langle \Psi | \hat{\mathcal{H}} | \Psi \rangle}{\langle \Psi | \Psi \rangle} = \frac{\int \Psi^*(\mathbf{R}; \theta) \hat{\mathcal{H}} \Psi(\mathbf{R}; \theta) d\mathbf{R}}{\int \Psi^*(\mathbf{R}; \theta) \Psi(\mathbf{R}; \theta) d\mathbf{R}}. \quad (9)$$

Monte Carlo integration is used to evaluate the high-dimensional integrals, sampling particle configurations $\mathbf{R}_i$ from the probability density $p \propto \Psi^2$ using the Metropolis–Hastings algorithm.³² The expectation value of the energy and its gradient are computed using the following Monte Carlo estimators:²⁷

$$\begin{aligned} \langle \hat{\mathcal{H}} \rangle_\theta &= \mathbb{E}_{\mathbf{R} \sim p}[E_L(\mathbf{R}; \theta)], \\ \nabla_\theta \langle \hat{\mathcal{H}} \rangle_\theta &= 2\mathbb{E}_{\mathbf{R} \sim p}[(E_L(\mathbf{R}; \theta) - \langle \hat{\mathcal{H}} \rangle_\theta) \nabla_\theta \log |\Psi(\mathbf{R}; \theta)|], \end{aligned} \quad (10)$$

where the symbol $\mathbb{E}_{\mathbf{R} \sim p}$ indicates an expectation value over points $\mathbf{R}$ sampled from the probability distribution, p , and $E_L(\mathbf{R}; \theta) = \hat{\mathcal{H}}\Psi(\mathbf{R}; \theta)/\Psi(\mathbf{R}; \theta)$ is the local energy. The accuracy of the VMC method depends on the accuracy with which the trial wavefunction is able to approximate the true ground state.

The neural VMC approach is distinguished from traditional applications of VMC by its use of a neural network as the trial wave function. The network inputs are the particle positions, the output is the value of the wavefunction for those inputs, and the variational parameters are the network weights and biases. Neural networks are such flexible and accurate function approximators that the accuracy of this approach rivals and sometimes exceeds that of traditional quantum chemical methods such as CCSD(T). The neural wavefunctions used in this work are based on the Psiformer architecture,²³ with the following functional form:

$$\Psi(\mathbf{R}) = e^{J(\mathbf{R})} \sum_k \prod_\chi \det[\phi_i^{k\chi}(\mathbf{r}_j^\chi; \{\mathbf{r}_{jj}^\chi\}; \{\mathbf{r}_j^\chi\})], \quad (11)$$

where $\chi \in \{\mu, e\}$ indexes the particle species. The wavefunction is a sum of determinants, indexed by the integer k , containing many-particle orbitals, $\phi_i^{k\chi}(\mathbf{r}_j^\chi; \{\mathbf{r}_{jj}^\chi\}; \{\mathbf{r}_j^\chi\})$, which are computed by a transformer-based neural network. The arguments of $\phi_i^{k\chi}$ are the position $\mathbf{r}_j^\chi$ of particle j of species χ , the position of every other particle of the same species, $\{\mathbf{r}_{jj}^\chi\}$, and the position of every particle not of that species, $\{\mathbf{r}_j^\chi\}$. The position vectors $\mathbf{r}_j^\chi$ are ordered such that j iterates through all spin-up particles, followed by all spin-down particles for particle species χ . The wavefunction also contains an exponential Jastrow factor, $J(\mathbf{R})$, represented by an additional small neural network, which aids in representing short-range particle–particle correlations. By construction, the wavefunction is anti-symmetric with respect to the exchange of identical

particles, ensuring the Pauli exclusion principle is satisfied.²³ Further details of the neural network architecture are found in [Appendix A](#) and Refs. [12](#), [23](#), and [27](#).

To compute the hyperfine constant, we estimate the spin-resolved system-averaged pair density function $\bar{\rho}_{\sigma_{\mu}\sigma_e}(r)$ using Monte Carlo integration. The hyperfine constant is proportional to the value at the origin of the translationally averaged muon–electron spin density, $\Delta\rho(0) = \bar{\rho}_{\uparrow\uparrow}(0) - \bar{\rho}_{\uparrow\downarrow}(0)$. We find $\Delta\rho(0)$ by fitting the logarithm $\ln|\Delta\rho(r)|$ as a quadratic polynomial constrained to obey the generalized Kato cusp condition between two particles.³⁴ Further details are provided in [Appendix B](#).

III. RESULTS

We study two muoniated systems: the muoniated methyl radical (CH_2Mu), which contains 1 muon and 9 electrons, and the muoniated ethyl radical ($\text{C}_2\text{H}_4\text{Mu}$), which contains 1 muon and 17 electrons. For each system, we calculated the hyperfine constant at different levels of approximation, which we refer to as the clamped muon, quantum muon, and full quantum treatments:

Clamped Muon: The muon, hydrogen nuclei, and carbon nuclei are clamped at fixed positions within the Born–Oppenheimer approximation, with no treatment of mass effects. This is the level of approximation employed in most DFT calculations of hyperfine constants (which in addition approximate the electron–electron correlations using an effective exchange–correlation potential) and is most directly comparable with standard DFT results.

Quantum Muon: The muon is treated as a quantum particle and included, together with the electrons, in the ground-state wavefunction, while the hydrogen nuclei and carbon nuclei remain clamped. This allows for a more accurate evaluation of the hyperfine constant by explicitly sampling the quantum pair-density between the muon and the electrons.

Full Quantum (muoniated methyl radical only): The muon and the hydrogen nuclei are treated as quantum particles in the ground-state wavefunction, and only the carbon nucleus is clamped. This enables a complete study of the hyperfine constant that includes the perturbation of the hydrogen nuclei due to the implanted muon.

To position the clamped nuclei, we use DFT-computed geometries from the NIST chemistry WebBook.³⁵ These geometries were optimized within the Born–Oppenheimer approximation for the unsubstituted radicals, so no nuclear mass information enters the calculation and no account is taken of the influence of the light, delocalized muon on the equilibrium positions of the surrounding nuclei. When comparing predicted and experimental values, only the magnitude of the hyperfine constant is relevant.¹ The relative orientations of the spins of the muon and the unpaired electron in the radical do not affect the ground-state wavefunction of the Coulomb Hamiltonian, which excludes the hyperfine interaction and for which the latter is treated as a perturbation. This allows us to choose both the muon and the unpaired electron to be spin up when solving the Coulomb problem to obtain the unperturbed ground-state wavefunction.

As a validation of our methodology, we compute the properties of the muonium atom in vacuum. The quantum muon

approach yields an isotropic hyperfine constant of 4469(7) MHz and ground-state energy of $-0.497\,57(3)$ Hartree, accurately reproducing the established theoretical and experimental values of 4463 MHz and $-0.497\,59$ Hartree, respectively.^{36,37} Treating the muon as a clamped nucleus results in an energy of -0.5 Hartree and a hyperfine constant of 4529 MHz.³⁸ For muonium, the clamped-muon result is brought into agreement with the finite-mass value by scaling with the cube of the reduced mass, $A_{\text{Mu}} = (\mu_{\mu e}/m_e)^3 A_{\text{clamped}} = 4464$ MHz, which is exact in the nonrelativistic Schrödinger picture. The quantum muon Psiformer calculation captures this correction directly from the many-body wavefunction, which is essential for the molecular systems considered in the following where no analogous reduction is available.

A. Muoniated methyl radical

The results of our calculations of the hyperfine constant of the muoniated methyl radical are compared to those of DFT and experiment presented in [Table I](#). The magnitude of the clamped-muon hyperfine constant deviates from the experimental magnitude by 27(3) MHz (13%); this deviation is reduced to 22(3) MHz (11%) for the quantum-muon calculation and to 17(3) MHz (8%) for the full quantum calculation. (Note that the experiments do not determine the *sign* of the hyperfine constant.) Furthermore, the clamped-muon value is 40(3) MHz (21%) larger in magnitude than the DFT prediction despite describing the same underlying geometry; the origin of this discrepancy is discussed further in [Sec. IV B](#).

The corresponding ground-state energies are reported alongside DFT results in [Table II](#). The agreement between the DFT and clamped-muon Psiformer system energies is good, although DFT energies are not variational and may lie below the exact ground-state energy. As more particles are treated quantum mechanically, the system becomes less bound, consistent with the additional zero-point energy introduced by quantum motion.³⁹ The full quantum Psiformer simulations yield hyperfine constants closest to experiment, reflecting the more realistic description of the perturbation of the hydrogen nuclei in the presence of a quantum-mechanical muon. Although the DFT hyperfine constant appears surprisingly accurate, [Sec. IV A](#) shows that it is too small when environmental effects are taken into account.

[Figure 1](#) shows planar projections of the muon and electron spin densities when the muon is treated quantum mechanically. The unpaired electron occupies a *p*-like orbital centered on the carbon, but it is the very small negative electron spin density close to the plane of the molecule near the delocalized muon—hard to discern from the figure alone—which determines the sign of the calculated muon hyperfine constant.

TABLE I. Isotropic muon hyperfine constants for the muoniated methyl radical computed with Psiformer and DFT, compared with experimental data at 73 K. The brackets denote uncertainties in the last digit, where reported.

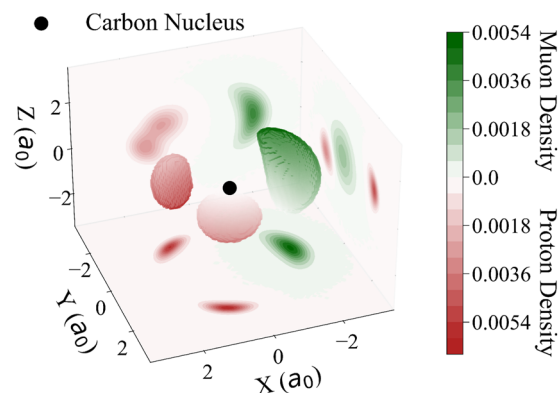
Muoniated methyl radical	Hyperfine constant A_{μ} (MHz)
Psiformer clamped muon	−228(3)
Psiformer quantum muon	−223(3)
Psiformer full quantum	−218(3)
DFT B3LYP/cc-pVDZ ⁴⁰	−188
Experiment (73 K) ¹⁹	±201.30(10)

TABLE II. Ground-state energies for the muoniated methyl radical computed with Psi-former and DFT. The brackets indicate the standard error on the mean of the last digit, where reported.

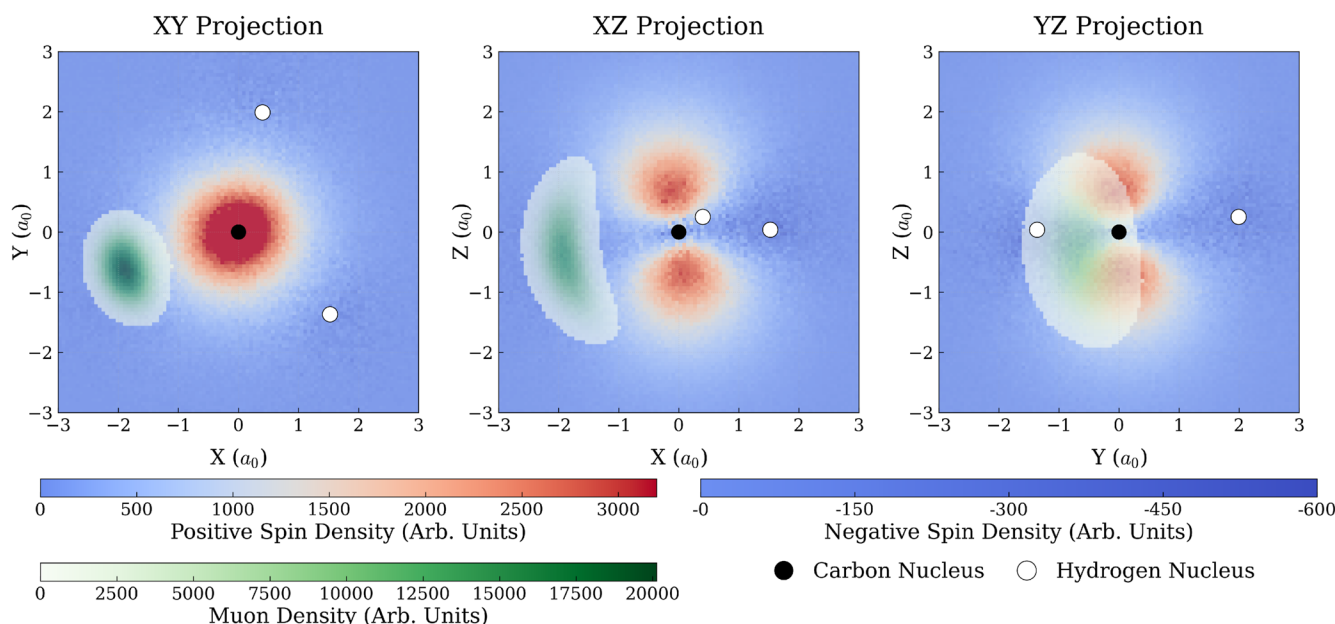
Muoniated methyl radical	Ground state energy (E_h)
Clamped muon	-39.8345(1)
Quantum muon	-39.8012(1)
Full quantum	-39.7814(2)
DFT NIST B3LYP/6-31G* ⁴¹	-39.8383

The need to treat the muon quantum mechanically is illustrated more clearly in Fig. 2: the muon is noticeably more delocalized than the protons when both are treated quantum mechanically, indicating that the Born–Oppenheimer approximation used in standard DFT may be insufficient to capture the muon’s quantum nature.

Figure 3 shows the sampled translationally averaged muon–electron spin density $\Delta\rho(r)$ as a function of the separation r between the electron and muon. For each system, the sampled electron–muon spacings are placed into radial bins of width $0.005a_0$ and the number of samples of each spin in each bin is divided by the bin volume to obtain an estimate of the sample density and thus the muon–electron spin density at that separation. Every point shows the result of this calculation for one bin in one system. The muon–electron spin density is extrapolated to $r = 0$ using a quadratic fit in the log domain, the parameters of which are chosen to ensure the Kato cusp condition is satisfied exactly. The

**FIG. 2.** 10% isosurfaces of the muon and proton densities in the ground state of the muoniated methyl radical (CH_2Mu) obtained from a full quantum Psi-former simulation. The proton density is indicated in red and the muon density in green. The densities were accumulated in $5.12 \times 10^{-4} a_0^3$ cubic bins and smoothed using a Gaussian convolution with $\sigma = 0.06a_0$. Contour projections of the binned function are shown on the axes. The quantum muon is more delocalized than the protons due to its lower mass.

extrapolated value and its statistical error determine the isotropic muon hyperfine constant A_μ and its uncertainty. See Appendix B for details of the data analysis and fitting procedure. All three systems produce similar values for $\Delta\rho(0)$, although the differences between them are greater than the fitting errors.

**FIG. 1.** Electron spin density and muon density of the muoniated methyl radical (CH_2Mu) with a quantum muon, projected into the XY, XZ, and YZ planes. The molecular structure is largely coplanar with the XY plane. The magnitude and sign of the electron spin density are indicated by blue–red colors, while the magnitude of the muon density is indicated by green. The electronic spin density is concentrated near the carbon atom, while the muon density is delocalized and centered some distance away.

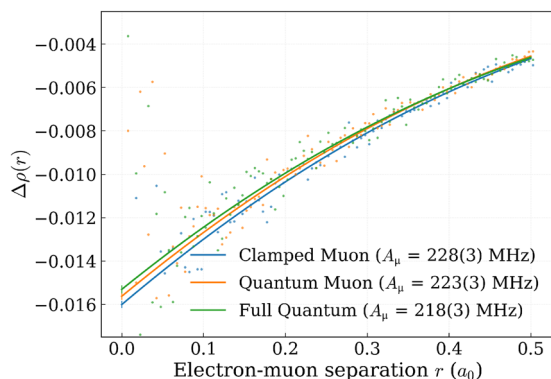


FIG. 3. Translationally averaged muon–electron spin density for the muoniated methyl radical (CH_2Mu) in the clamped muon, quantum muon, and full quantum systems. The muon–electron spin density is extrapolated to obtain $\Delta\rho(r=0)$ using a quadratic fit in the log domain. The extrapolated value and its statistical error determine the isotropic muon hyperfine constant A_μ and its uncertainty.

B. Muoniated ethyl radical

The results of our calculations of the hyperfine constant of the muoniated ethyl radical are compared with DFT and the experiment in Table III. The quantum-muon hyperfine constant is consistent with the experiment and 71(6) MHz (15%) larger in magnitude than the hyperfine constant calculated with a clamped muon. Furthermore, the clamped-muon hyperfine constant is smaller in magnitude than the DFT predictions by 13(4) MHz (3%) or 28(4) MHz (6%), depending on the DFT basis set.

As shown in Table IV, the clamped-muon ground-state energies computed using DFT and the Psiformer are in reasonably good agreement. By contrast, when the muon is treated quantum mechanically, the Psiformer energies are less bound, reflecting the additional muon zero-point energy, as in the methyl radical case.

Figure 4 shows planar projections of the muon and electron spin densities for the quantum-muon system. The hyperfine constant is positive because the electron spin density at the muon site, arising from the tail of the singly occupied p orbital on the carbon atom opposite the muon, is itself positive. This indicates that in the

TABLE III. Isotropic muon hyperfine constants of the muoniated ethyl radical computed with Psiformer and DFT, compared with experimental results. The brackets denote uncertainties in the last digit, where reported.

Muoniated ethyl radical	Hyperfine constant A_μ (MHz)
Psiformer clamped muon	466(4)
Psiformer quantum muon	537(4)
DFT O3LYP/6-31G(d,p) ^{a,10}	478.9
DFT O3LYP/Chipman ^{a,10}	493.8
Experiment (5–6 K) ^{b,42}	530(3), 535(2), 542(2)

^aThe two DFT results use different basis sets: 6-31G(d,p) and Chipman. The Chipman basis is more accurate as it is designed for Fermi contact spin density calculations.⁴³

^bThe muoniated ethyl radical is created in an environment of supercages of USY, HY, and NaY, silicate minerals with porous crystalline structures.^{42,44}

TABLE IV. Ground-state energies for the muoniated ethyl radical computed with Psiformer and DFT. The brackets indicate the standard error on the mean of the last digit, where reported.

Muoniated ethyl radical	Ground state energy (E_h)
Psiformer clamped muon	−79.148 62(5)
Psiformer quantum muon	−79.112 8(2)
NIST DFT B3LYP/6-31G* ⁴⁵	−79.157 9

muoniated ethyl radical, the hyperfine constant is dominated by the spin of the unpaired electron in the radical.

Figure 5 shows the sampled translationally averaged muon–electron spin density, $\Delta\rho(r)$, accumulated using radial bins of width $0.005a_0$ and extrapolated to $r=0$ exactly as for the muoniated methyl radical. The difference between the quantum-muon and clamped-muon calculations can be interpreted as a consequence of muon zero-point motion, which shifts the muon probability density into regions of higher positive spin density.

IV. DISCUSSION AND CONCLUSIONS

A. Temperature and environmental impacts

In both the Psiformer and DFT calculations, the hyperfine constant is obtained from ground-state wavefunctions in vacuum at zero temperature. However, temperature and the medium surrounding the muoniated radical can introduce systematic effects that influence experimentally measured hyperfine constants.^{10,44} We, therefore, compare primarily with the lowest-temperature available measurements, for which thermal effects should be smallest, although condensed-phase experiments may still contain residual anisotropic and environmental contributions that are not included in the present model.

For the muoniated methyl radical, experimental measurements are available down to 73 K in a liquid ketene environment.¹⁹ The effect of temperature on the muon hyperfine constant is expected to be small (on the order of 1 MHz) as the results of measurements between 73 and 223 K vary by no more than 0.34 MHz (0.17%) and proton hyperfine constants measured between 4 and 96 K differ by less than 0.3%.¹⁹ These observations are consistent with the absence of low-energy vibrational modes that would affect the hyperfine constant below 73 K.

However, proton hyperfine constants for the methyl radical in vacuum at 25 K are larger than in any of the other 11 environments explored by McKenzie *et al.*,¹⁹ irrespective of temperature. We, therefore, expect the liquid ketene environment to reduce the magnitude of the proton hyperfine constant relative to the vacuum value. By analogy, one would also expect the muon hyperfine constant to reduce. Taking the lowest-temperature measurement for each environment measured by McKenzie *et al.*,¹⁹ the average magnitude of the proton hyperfine constant in methyl is reduced by 1.38 ± 1.15 MHz relative to its vacuum value at 25 K, where the uncertainty includes the standard deviation across environments. This corresponds to a fractional reduction of about 2.2% when moving from vacuum to an average environment.

Assuming an analogous environmental effect for the muon, we can estimate the vacuum value of the muon hyperfine constant

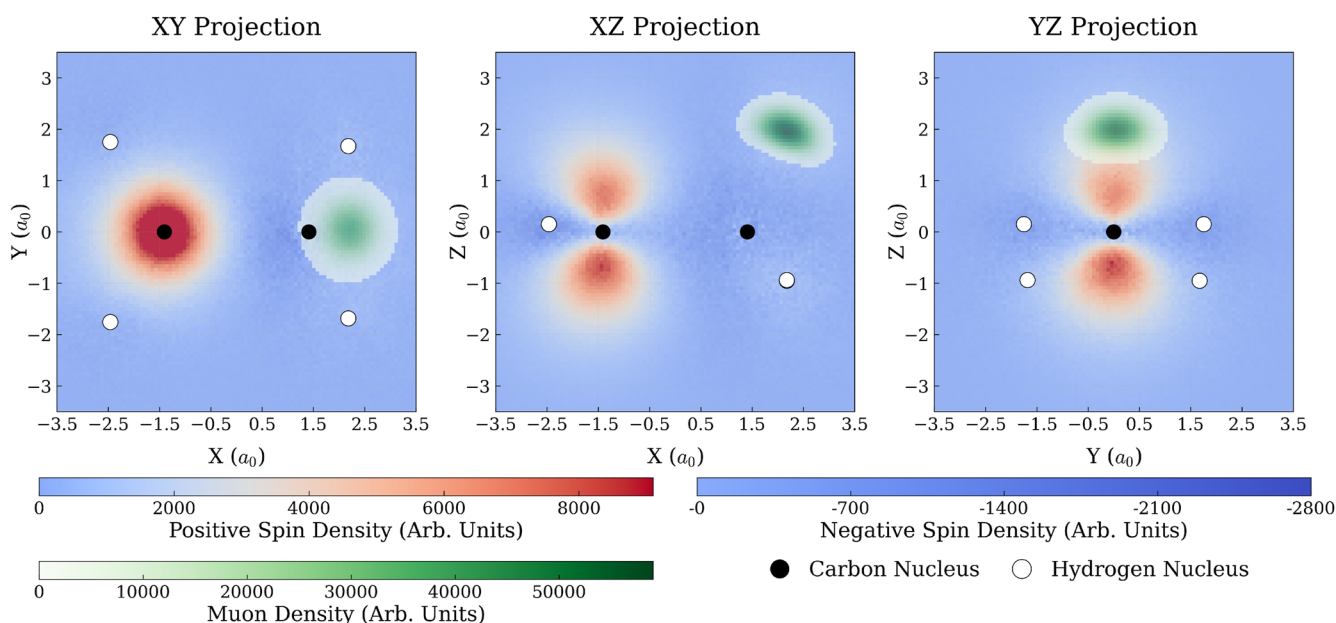


FIG. 4. Electron spin density and muon density of the muoniated ethyl radical (C_2H_4Mu) with a quantum muon, projected into the XY, XZ, and YZ planes. The magnitude and sign of the spin density are indicated by the blue-red colors while the magnitude of the muon density is indicated by green. The electronic spin density is concentrated near the carbon atom at $X \approx -1.5a_0$, while the muon density is delocalized some distance from the other carbon atom at $X \approx 1.5a_0$.

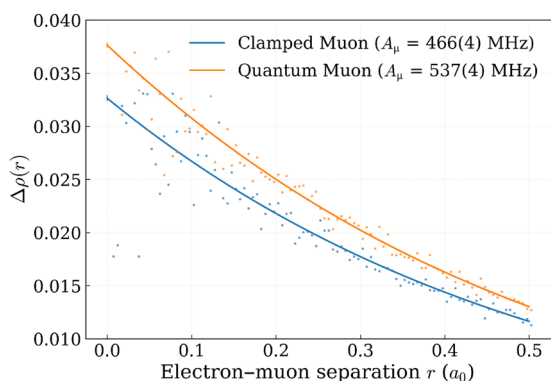


FIG. 5. Translationally averaged muon-electron spin densities of the muoniated ethyl radical (C_2H_4Mu) in the clamped muon and quantum muon systems. The muon-electron spin density is extrapolated to obtain $\Delta\rho(r=0)$ using a quadratic fit in the log domain. The extrapolated value and its statistical error determine the isotropic muon hyperfine constant A_μ and its uncertainty.

by increasing the magnitude of the value in a ketene environment by 2.2%, from 201.30(10) MHz to about 205.6 ± 3.6 MHz. (Since the muon is more delocalized than a proton and likely to be more strongly affected by the environment, the true environmental correction is probably larger.) With this correction, the deviation between experiment and the quantum-muon Psiformer result is reduced from 22(3) MHz (11%) to 17(5) MHz (8%), while the result

of the full-quantum Psiformer calculation differs from experiment by only 12(5) MHz (6%). The error in the DFT value increases to about 18(4) MHz (9%).

The experimental measurements for the muoniated ethyl radical were carried out at 5–6 K in silicate mineral environments, which are also expected to affect the muon hyperfine constant.⁴² The systematic impact of the silicate host can be estimated by comparing room-temperature measurements in these hosts with gas phase data. At 298 K, the gas phase muon hyperfine constant is 329.5(4) MHz,⁴⁶ whereas at 300 K in NaY, HY, and USY silicate minerals, it is 349(1), 346(1), and 340(1) MHz, respectively.⁴² The silicate environment increases the hyperfine constant by 16(4) MHz, on average. Experiments also show strong temperature dependence, making it difficult to correct simultaneously for both environmental and thermal effects; however, at 300 K, the silicate-induced shift is small compared to the overall temperature dependence. Reference 42 attributes this temperature dependence to binding at cation sites and framework OH groups, mechanisms expected to strengthen at lower temperatures. As a simple approximation, we nonetheless assume that the temperature variation of the environmental offset is small in comparison with the offset itself and subtract the 16 MHz environmental offset from the low-temperature silicate data to estimate a corresponding vacuum value, recognizing that this probably underestimates the matrix effect at 5 K. Averaging the 5–6 K measurements in NaY, HY, and USY and applying this shift yields an approximate low temperature *vacuum* muon hyperfine constant of 520(5) MHz. The closest DFT prediction underestimates this value by 36(5) MHz (7%), whereas the quantum muon calculation overestimates it by about 17(6) MHz (3%).

For both the methyl and ethyl muoniated radicals, the quantum Psiformer calculations of muon hyperfine constants lie closest to the estimated low-temperature experimental results after environmental and temperature corrections have been made. However, the need for these approximate corrections limits how directly we can validate the method. An alternative would be to explore low-temperature solids, where such corrections would not be required. Experimental data and DFT-based predictions exist for muonium in silicon,^{47,48} the insulators NaF and CaF₂, and diamond.²² Periodic neural wavefunction calculations on small supercells should be tractable for such systems, providing a direct test of the impact of quantum-mechanical muon delocalization on the hyperfine constants in solids and a benchmark against two-component DFT. However, it should be noted that in addition to the *interstitial muonium* isotropic state, solid-state systems sometimes also exhibit anisotropic muonium states such as *bond-centered muonium*, so the model would have to include anisotropic effects.

B. Comparison with density functional theory

Significant deviations exist between Psiformer clamped muon and DFT calculations of the muon hyperfine constants of muoniated methyl and ethyl radicals despite the fact that both approaches clamp the muon at a fixed position. In the absence of methodological errors, the two methods would be expected to yield the same hyperfine constant for a given geometry. In DFT, however, the choice of approximate exchange–correlation functional and basis set can also affect the calculated hyperfine constants, as seen for the muoniated ethyl radical presented in Table III. By contrast, Psiformer employs an *ab initio* variational method that has been shown to represent interacting many-body wavefunctions accurately and to yield highly accurate ground-state energies across a range of systems.^{12,23} It is, therefore, likely that the discrepancies between the clamped-muon hyperfine constants arise primarily from the known limitations of DFT, rather than from a failing of the neural-wavefunction approach. Nonetheless, within these limitations and approximations, the results also illustrate the surprisingly robust performance of DFT.

C. Limitations

The main limitation of the neural wavefunction method is its cost, which scales as $O(N^{3-4})$ for N order of several dozen particles. This is not bad in comparison with most other many-body methods:²³ the cost of a full configuration interaction (FCI) calculation increases exponentially with N ;⁴⁹ while standard implementations of the highly accurate CCSD(T) method scale as $O(N^7)$.²³ However, the pre-factor of the scaling law that determines the cost of a neural wavefunction simulation is large enough to make it impractical to simulate systems with more than around 100 electrons at present.

The cost of conventional implementations of DFT scales as $O(N^3)$. This is not very different from the scaling of neural wavefunction methods, but the DFT prefactor is vastly smaller. For the systems considered here, DFT simulations can be run in seconds on suitable high-performance computing hardware.⁵⁰ The most demanding Psiformer calculations performed here for the full quantum muoniated methyl radical and the muoniated ethyl

radical with a quantum muon — required around seven days on four NVIDIA A100 Tensor Core graphics processing units (GPUs). The relative speed of DFT allows nuclear geometries to be relaxed to find optimized structures, which is usually impractical with slower many-body approaches. A hybrid strategy, in which geometries are optimized with DFT and hyperfine constants are evaluated with a neural-wavefunction method, may prove effective in future studies.

In the full quantum calculation for the muoniated methyl radical, the proton positions change in response to the presence of the quantum muon, while the carbon nucleus remains clamped at its DFT-optimized position. We did not conduct a full quantum calculation for ethyl here, although such a calculation has since been reported by another group.⁵¹ For future molecules, we could treat all protons quantum mechanically alongside the muon, with the fixed nuclear parameters—the carbon–carbon distance in the case of the muoniated ethyl radical—optimized between runs to minimize the system energy. Such an approach would be scalable and would also benefit from the more accurate quantum description of the protons. In addition, the relaxed geometry could, in principle, be determined using a force-based approach similar to that discussed in Ref. 52. We expect that this approach would yield even more accurate hyperfine constants in future calculations, and comparing such relaxed geometries with the DFT geometries would directly quantify the influence of the delocalized quantum muon on the equilibrium positions of the surrounding nuclei.

Another limitation of the approach is the statistical uncertainty in the calculation of the hyperfine constant. This arises from the limited number of samples accumulated at small electron–muon separations, which leads to the relatively large errors in the on-top translationally averaged muon–electron spin density and the sample deviations observed in Figs. 3 and 5. This error was mitigated by using a physically motivated fit function constrained by analytical cusp conditions and performing the fit in the logarithmic domain with weights inversely proportional to the variance to reduce the impact of samples with high variance. Remaining errors could be further reduced by adopting more sophisticated sampling schemes that better resolve the short-range region.

ACKNOWLEDGMENTS

We acknowledge the EUROfusion consortium for providing computing time on the Leonardo Supercomputer at CINECA in Bologna. The EUROfusion Consortium was funded by the European Union via the Euratom Research and Training Programme (Grant Agreement No. 101052200—EUROfusion). Views and opinions expressed are, however, those of the authors only and do not necessarily reflect those of the European Union or the European Commission; neither the European Union nor the European Commission can be held responsible for them. We also gratefully acknowledge the Gauss Centre for Supercomputing e.V. for funding this project by providing computing time through the John von Neumann Institute for Computing (NIC) on the GCS Supercomputer JUWELS at Jülich Supercomputing Centre. A.P.F.'s Ph.D. work was supported by the UK Engineering and Physical Sciences Research Council under Grant No. EP/W524323/1. Finally, we thank Jarvist Frost, Adrian Hiller, Francis Pratt, and Leandro Liborio for helpful discussions.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jamie Carr: Conceptualization (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Mathias Volkai:** Conceptualization (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **W. M. C. Foulkes:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal). **Andres Perez Fadon:** Conceptualization (equal); Investigation (equal); Methodology (equal); Project administration (equal); Software (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.19089248>.⁵⁴

APPENDIX A: PSIFORMER IMPLEMENTATION AND HYPERPARAMETERS

All Psiformer calculations were performed using one node and four NVIDIA A100 Tensor Core GPUs of the Booster Module of the Leonardo high-performance computer.

The calculations were carried out using a modified version of the publicly available FermiNet code,⁵³ extended to support wavefunctions over multiple particle species of arbitrary mass and charge, as first discussed in Ref. 27. Further modifications were made to accumulate the spin-resolved muon–electron pair-density histograms analyzed in Appendix B. These changes will be integrated into the main FermiNet repository to allow all the results reported here to be reproduced using public code. Supplementary information with data and implementation code can be found in Ref. 54.

The Psiformer hyperparameters used for all systems are listed in Table V.

APPENDIX B: ESTIMATION, FITTING, AND STATISTICS OF THE SPIN-RESOLVED SYSTEM-AVERAGED PAIR DENSITY FUNCTION

To compute the hyperfine constant, we estimate the spin-resolved system-averaged pair density function $\bar{\rho}_{\sigma_\mu\sigma_e}(r)$ by accumulating Monte Carlo statistics of particle positions for aligned and anti-aligned electron–muon spin arrangements. Using a histogram approach, we divide the space around the muon into concentric spherical shells, indexed by k , with inner radius r_k and outer radius r_{k+1} . For N_{samples} Monte Carlo configurations, $n_{\sigma_e}(k)$ counts the number of electrons with spin σ_e found within the k -th shell centered on the muon. The pair density for a given radial bin is approximated

TABLE V. Psiformer parameters for all systems studied. Learning rate is scheduled based on iteration number t .

Parameter	Value
Psiformer determinants count	16
Psiformer layers	2
Psiformer heads	4
Psiformer heads width	64
MLP hidden dims	256
Jastrow MLP layers	3
Jastrow MLP width	64
Learning rate	$(0.01 + \frac{0.01t}{10\,000})^{-1}$
Local energy clipping	5.0
KFAC momentum	0
KFAC covariance moving average decay	0.95
KFAC norm constraint	10^{-3}
KFAC damping	10^{-3}
Steps between parameter updates	10
Walker batch size	4096
Initial walker width	1.0
Initial move width	0.1

by normalizing this count by the shell volume and the total number of samples,

$$\bar{\rho}_{\sigma_\mu\sigma_e}(r_k) \approx \frac{n_{\sigma_e}(k)}{\frac{4}{3}\pi(r_{k+1}^3 - r_k^3)N_{\text{samples}}}. \quad (\text{B1})$$

When the muon is clamped, the fixed location of the muon is used as the center of the spherical shells. In contrast, for quantum muon simulations, the muon position within each Monte Carlo sample becomes the center, so that the shell radius is the electron–muon spacing.

The hyperfine constant is proportional to $\Delta\rho(0) = \bar{\rho}_{\uparrow\uparrow}(0) - \bar{\rho}_{\uparrow\downarrow}(0)$. However, the vanishing volume of the spherical shells as $r \rightarrow 0$ results in low sample counts and characteristic Poisson noise in estimates of $\bar{\rho}_{\sigma_\mu\sigma_e}(r)$ when r is small. It is computationally infeasible to generate more samples for this region, so this is addressed by fitting the translationally averaged muon–electron spin density $\Delta\rho(r)$ in the short-range region ($r < 0.5a_0$) and extrapolating to zero. Because the electron–muon pair density exhibits a cusp-like singularity at coalescence,⁵⁵ we expect $\Delta\rho(r)$ to exhibit exponential behavior at small r . Accordingly, we employ a quadratic fit to the logarithm of the translationally averaged muon–electron spin density,

$$\ln|\Delta\rho(r)| = a + br + cr^2, \quad (\text{B2})$$

where a , b , and c are fitting parameters. The intercept a yields $\ln\Delta\rho(0)$, from which $\Delta\rho(0)$ is recovered.

We constrain the fit by applying the generalized Kato's cusp condition,³⁴ which describes how the many-body wavefunction behaves as any two distinguishable charged particles approach one another. In the case of a muon and an electron, this reads

$$\left.\frac{d\bar{\rho}}{dr}\right|_{r=0} = 2\mu_{\mu e}q_{\mu}q_e\bar{\rho}(0), \quad (\text{B3})$$

where $\mu_{\mu e}$ is the reduced mass of the muon–electron pair and q_{μ}, q_e are their charges, in atomic units. Applying this condition to the translationally averaged muon–electron spin density, $\Delta\rho(r)$, fixes the linear coefficient b in the logarithmic fitting ansatz,

$$\left. \frac{d}{dr} \ln \Delta\rho(r) \right|_{r=0} = b = -2\mu_{\mu e}, \quad (\text{B4})$$

thereby reducing the number of free fitting parameters. A weighted least-squares fit is employed for the remaining parameters, using weights inversely proportional to the variance of each sample. The reported uncertainty in the hyperfine constant is obtained by propagating the standard error of the fitted intercept parameter, a , derived from the covariance matrix of the fit. The jackknife and bootstrap procedures⁵⁶ were also used to estimate the uncertainty in the intercept parameter. All three methods gave consistent results, yielding uncertainties of ~ 3 and 5 MHz in the muoniated methyl and ethyl hyperfine constants, respectively.

Two validity checks were employed for all results. First, we verified that the integral of $\Delta\rho(r)$ was unity to within floating-point error, consistent with the presence of a single unpaired electron in each system. Second, although the fit for $\Delta\rho(r)$ explicitly imposes the Kato cusp condition, the neural network does not. The high quality of the resulting fits suggests that the neural network has learned an accurate representation of the cusp.

REFERENCES

- 1A. Amato and E. Morenzoni, *Introduction to Muon Spin Spectroscopy: Applications to Solid State and Material Sciences, Lecture Notes in Physics Vol. 961* (Springer International Publishing, 2024).
- 2C. J. Rhodes, “Muonium—the second radioisotope of hydrogen: A remarkable and unique radiotracer in the chemical, materials, biological and environmental sciences,” *Sci. Prog.* **95**, 101–174 (2012).
- 3*Muon Spectroscopy: An Introduction*, 1st ed., edited by S. J. Blundell, R. De Renzi, T. Lancaster, and F. L. Pratt (Oxford University Press, Oxford, 2021).
- 4A. D. Hillier, S. J. Blundell, I. McKenzie, I. Umegaki, L. Shu, J. A. Wright, T. Prokscha, F. Bert, K. Shimomura, A. Berlie, H. Alberto, and I. Watanabe, “Muon spin spectroscopy,” *Nat. Rev. Methods Primers* **2**, 4 (2022).
- 5I. McKenzie, “The positive muon and μ SR spectroscopy: Powerful tools for investigating the structure and dynamics of free radicals and spin probes in complex systems,” *Annu. Rep. Prog. Chem., Sect. C: Phys. Chem.* **109**, 65–112 (2013).
- 6H. T. G. Walsgrove, P. W. Percival, and D. P. Gates, “Probing radical addition to 1-phosphabutadienes by employing muonium as a ‘light isotope’ of hydrogen,” *Chem. Eur. J.* **30**, e203202869 (2024).
- 7E. Roduner and H. Fischer, “Muonium substituted organic free radicals in liquids. Theory and analysis of μ SR spectra,” *Chem. Phys.* **54**, 261–276 (1981).
- 8E. Roduner, P. W. Percival, D. G. Fleming, J. Hochmann, and H. Fischer, “Muonium-substituted transient radicals observed by muon spin rotation,” *Chem. Phys. Lett.* **57**, 37–40 (1978).
- 9S. L. Thomas and I. Carmichael, “Hyperfine interactions in muonium-containing radicals,” *Physica B* **374–375**, 290–294 (2006).
- 10K. Yamada, Y. Kawashima, and M. Tachikawa, “Accurate prediction of hyperfine coupling constants in muoniated and hydrogenated ethyl radicals: *Ab initio* path integral simulation study with density functional theory method,” *J. Chem. Theory Comput.* **10**, 2005–2015 (2014).
- 11J. Peck, F. Pratt, and S. Cottrell, “Simulating hyperfine coupling constants of muoniated radicals using density functional theory calculations,” Technical Report No. RAL-TR-2015-013, STFC Rutherford Appleton Laboratory, 2015.
- 12D. Pfau, J. S. Spencer, A. G. D. G. Matthews, and W. M. C. Foulkes, “*Ab initio* solution of the many-electron Schrödinger equation with deep neural networks,” *Phys. Rev. Res.* **2**, 033429 (2020).
- 13S. J. Blundell and T. Lancaster, “DFT + μ : Density functional theory for muon site determination,” *Appl. Phys. Rev.* **10**, 021316 (2023).
- 14J. P. Perdew and S. Kurth, “Density functionals for non-relativistic coulomb systems in the new century,” in *A Primer in Density Functional Theory*, edited by C. Fiolhais, F. Nogueira, and M. A. L. Marques (Springer, Berlin, Heidelberg, 2003), pp. 1–55.
- 15T. A. Claxton, A. M. Graham, S. F. J. Cox, D. M. Maric, P. F. Meier, and S. Vogel, “Vibrationally averaged spin densities on muons and protons in the hydroxyl and ethyl radicals,” *Hyperfine Interact.* **65**, 913–926 (1991).
- 16T. A. Claxton and A. M. Graham, “A chemical interpretation of vibrationally induced barriers to hindered internal rotation,” *J. Chem. Soc., Faraday Trans. 2* **83**, 2307–2317 (1987).
- 17M. C. Böhm, R. Ramírez, and J. Schulte, “Finite-temperature properties of the muonium substituted ethyl radical CH_2MuCH_2 : Nuclear degrees of freedom and hyperfine splitting constants,” *Mol. Phys.* **103**, 2407–2436 (2005).
- 18Y. K. Chen, D. G. Fleming, and Y. A. Wang, “Theoretical calculations of hyperfine coupling constants for muoniated butyl radicals,” *J. Phys. Chem. A* **115**, 2765–2777 (2011).
- 19I. McKenzie, J.-C. Brodovitch, K. Ghandi, B. M. McCollum, and P. W. Percival, “Hyperfine coupling in methyl radical isotopomers,” *J. Phys. Chem. A* **111**, 10625–10634 (2007).
- 20E. Boroński and R. M. Nieminen, “Electron-positron density-functional theory,” *Phys. Rev. B* **34**, 3820–3831 (1986).
- 21M. Goli and S. Shahbazian, “Two-component density functional theory for muonic molecules: Inclusion of the electron-positive muon correlation functional,” *J. Chem. Phys.* **156**, 044104 (2022).
- 22L. Deng, Y. Yuan, F. L. Pratt, W. Zhang, Z. Pan, and B. Ye, “Two-component density functional theory study of quantized muons in solids,” *Phys. Rev. B* **107**, 094433 (2023).
- 23I. v. Glehn, J. S. Spencer, and D. Pfau, “A self-attention ansatz for *ab-initio* quantum chemistry,” *arXiv:2211.13672* [physics] (2023).
- 24J. Hermann, J. Spencer, K. Choo, A. Mezzacapo, W. M. C. Foulkes, D. Pfau, G. Carleo, and F. Noé, “*Ab initio* quantum chemistry with neural-network wavefunctions,” *Nat. Rev. Chem.* **7**, 692–709 (2023).
- 25K. Choo, A. Mezzacapo, and G. Carleo, “Fermionic neural-network states for *ab-initio* electronic structure,” *Nat. Commun.* **11**, 2368 (2020).
- 26J. Hermann, Z. Schätzle, and F. Noé, “Deep-neural-network solution of the electronic Schrödinger equation,” *Nat. Chem.* **12**, 891–897 (2020).
- 27G. Cassella, W. M. C. Foulkes, D. Pfau, and J. S. Spencer, “Neural network variational Monte Carlo for positronic chemistry,” *Nat. Commun.* **15**, 5214 (2024).
- 28D. Linteau, S. Moroni, G. Carleo, and M. Holzmann, “Neural wave functions for high-pressure atomic hydrogen,” *Phys. Rev. Res.* **8**, L022007 (2026).
- 29I. McKenzie, V. L. Karner, and R. Scheuermann, “Analysis of avoided level crossing muon spin resonance spectra of muoniated radicals in anisotropic environments: Estimation of muon dipolar hyperfine parameters for Lorentzian-like Δ_1 resonances,” *Quantum Beam Sci.* **8**(2), 15 (2024).
- 30Generalized from Amato.¹
- 31The hyperfine constant represents the energy shift due to the alignment or anti-alignment of the muon and system spins.
- 32W. M. C. Foulkes, L. Mitás, R. J. Needs, and G. Rajagopal, “Quantum Monte Carlo simulations of solids,” *Rev. Mod. Phys.* **73**, 33–83 (2001).
- 33J. Martens and R. Grosse, “Optimizing neural networks with Kronecker-factored approximate curvature,” *Proc. Mach. Learn. Res.* **37**, 2408–2417 (2015).
- 34R. T. Pack and W. B. Brown, “Cusp conditions for molecular wavefunctions,” *J. Chem. Phys.* **45**, 556–559 (1966).
- 35K. C. Hafner *et al.*, “Computed 3-D structures,” in *NIST Standard Reference Database 69* (National Institute of Standards and Technology, 2015), last revised 9 February 2015.
- 36M. I. Eides, “Hyperfine splitting in muonium: Accuracy of the theoretical prediction,” *Phys. Lett. B* **795**, 113–116 (2019).
- 37S. Kanda, Y. Fukao, Y. Ikeda, K. Ishida, M. Iwasaki, D. Kawall, N. Kawamura, K. M. Kojima, N. Kurosawa, Y. Matsuda, T. Mibe, Y. Miyake, S. Nishimura, N. Saito, Y. Sato, S. Seo, K. Shimomura, P. Strasser, K. S. Tanaka, T. Tanaka, H. A. Torii,

A. Toyoda, and Y. Ueno, "New precise spectroscopy of the hyperfine structure in muonium with a high-intensity pulsed muon beam," *Phys. Lett. B* **815**, 136154 (2021).

³⁸Obtained using the electron density at a hydrogen nucleus $\frac{1}{\pi a_0^3}$.

³⁹J. S. Möller, D. Ceresoli, T. Lancaster, N. Marzari, and S. J. Blundell, "Quantum states of muons in fluorides," *Phys. Rev. B* **87**, 121108 (2013).

⁴⁰F. Pratt, L. Liborio, and A. Hillier, "Anisotropic hyperfine constants in muon spin spectroscopy discussion," private communication, ISIS Neutron and Muon Source Muon Group (2025).

⁴¹P. Linstrom and W. Mallard, "Methyl radical (CH₃) computed chemical structure," in *NIST Chemistry WebBook, NIST Standard Reference Database Number 69* (National Institute of Standards and Technology, Gaithersburg, MD, 2025).

⁴²M. D. Bridges, D. J. Arseneau, D. G. Fleming, and K. Ghandi, "Hyperfine interactions and molecular motion of the mu-ethyl radical in faujasites: NaY, HY, and USY," *J. Phys. Chem. C* **111**, 9779–9793 (2007).

⁴³D. M. Chipman, "Gaussian basis sets for calculation of spin densities in first-row atoms," *Theor. Chim. Acta* **76**, 73–84 (1989).

⁴⁴K. Ghandi, F. E. Zahariev, and Y. A. Wang, "Theoretical studies of alkyl radicals in the NaY and HY zeolites," *J. Phys. Chem. A* **109**, 7242–7250 (2005).

⁴⁵P. Linstrom and W. Mallard, "Ethyl radical (C₂H₅) computed chemical structure," in *NIST Chemistry WebBook, NIST Standard Reference Database Number 69* (National Institute of Standards and Technology, Gaithersburg, MD, 2025).

⁴⁶P. W. Percival, J.-C. Brodovitch, S.-K. Leung, D. Yu, R. F. Kiefl, D. M. Garner, D. J. Arseneau, D. G. Fleming, A. Gonzalez, J. R. Kempton, M. Senba,

K. Venkateswaran, and S. F. J. Cox, "Hyperfine constants for the ethyl radical in the gas phase," *Chem. Phys. Lett.* **163**, 241–245 (1989).

⁴⁷E. Holzschuh, "Direct measurement of muonium hyperfine frequencies in Si and Ge," *Phys. Rev. B* **27**, 102–111 (1983).

⁴⁸H. Katayama-Yoshida and K. Shindo, "Hyperfine coupling constant of positive muon in silicon," *Phys. Rev. Lett.* **51**, 207–210 (1983).

⁴⁹G. H. Booth and A. Alavi, "Approaching chemical accuracy using full configuration-interaction quantum Monte Carlo: A study of ionization potentials," *J. Chem. Phys.* **132**, 174104 (2010).

⁵⁰N. Schuch and F. Verstraete, "Computational complexity of interacting electrons and fundamental limitations of density functional theory," *Nat. Phys.* **5**, 732–735 (2009).

⁵¹P. Cai, Y. Qian, L. Deng, W. Fu, L. Yang, Z. Sun, X.-Z. Li, E.-G. Wang, L. Chen, W. Ren, and J. Chen, "Permutation invariant multi-scale full quantum neural network wavefunction," [arXiv:2603.12233](https://arxiv.org/abs/2603.12233) [physics.chem-ph] (2026).

⁵²Y. Qian, X. Li, and J. Chen, "Force and stress calculations with a neural-network wave function for solids," *Faraday Discuss.* **254**, 529–541 (2024).

⁵³J. S. Spencer, D. Pfau, and FermiNet Contributors (2020). "FermiNet," GitHub. [http://github.com/deepmind/ferminet](https://github.com/deepmind/ferminet)

⁵⁴J. Carr, M. Volkai, W. M. C. Foulkes, and A. Perez Fadon (2026). "Supplementary information for 'Neural wavefunction calculations of μ SR isotropic hyperfine coupling constants with quantum muons and protons,'" Zenodo. <https://doi.org/10.5281/zenodo.19089248>

⁵⁵This cusp arises from the Coulomb singularity in the electron–muon interaction, analogous to Kato's cusp condition for electron–nucleus interactions.

⁵⁶E. Bradley, *The Jackknife, the Bootstrap and Other Resampling Plans, CBMS-NSF Regional Conference Series in Applied Mathematics Vol. 38* (Society for Industrial and Applied Mathematics, Philadelphia, PA, 1982).